

20-Day Lesson Plan: Stack-Neutral Backend & Web Application Development (*Java / Jakarta EE Focus*)

Day/ Session Unit	Topic	Java (Jakarta EE) Focus – Elaborated (This is BASLINE of the delivered topic)	Python (Flask) Equivalent (Will be discussed / students are to correlate if they know this OR sometime later)	Node.js (Express) Equivalent (Will be discussed / students are to correlate if they know this OR sometime later)
1	Intro to Dynamic Web Applications	Servlet containers, WAR deployment, HTTP request lifecycle, statelessness, TCP/IP substrate,	Flask app with app.route("/")	Express app.get("/")
2	HTTP Protocol Deep Dive	Idempotency, safe vs unsafe methods, header taxonomy, MIME types, status code ontology, servlet request/response contracts	Flask request/response objects	Express req / res
3	Jakarta EE Overview	Servlet API, JSP, MVC stratification, dependency injection (CDI), inversion of control, deployment descriptors	Flask WSGI app	Express middleware stack
4	Dynamic Rendering	JSP templating, JSTL tag libraries, EL (Expression Language), polymorphic rendering pipelines, server-side view composition	render_template with Jinja2	res.render() with template engine
5	Request & Response Lifecycle	Request dispatching, parameter binding, response buffering, I/O streams, servlet filters, request wrappers	Flask request context	Express routing lifecycle
6	Sessions & Cookies	Session affinity, cookie persistence, JSESSIONID, CSRF tokens, URL rewriting heuristics, session clustering	Flask session dict	Express express-session

7	Tiered Architectures (2-Tier & 3-Tier Combined)	2-Tier: monolithic coupling, direct JDBC invocation, transactional boundaries. 3-Tier: presentation tier, business logic encapsulation, persistence abstraction, decoupled orchestration, layered MVC design	Flask Blueprints + SQLAlchemy service layer	Express routers + Sequelize/Mongoose service modules
8	RDBMS Integration (MySQL)	Relational schema design, ACID compliance, JDBC API, connection pooling, ResultSet, SQL Injection, Stored Procedure,	Flask + SQLAlchemy ORM	Express + Sequelize ORM
9	schema less document DB Integration (MongoDB)	Document-oriented storage, BSON serialization, CAP theorem trade-offs, schema-less persistence, MongoDB Java driver	Flask + PyMongo	Express + Mongoose
10	HTTP API Design	Endpoint granularity, verb semantics, response codes taxonomy, payload serialization (JSON/XML), content negotiation. Using POSTMAN / cURL etc.	Flask-RESTful	Express REST routes
11	REST API Basics	JAX-RS endpoints, uniform interface constraint, resource identification, stateless interactions, hypermedia-driven architecture hypermedia controls (HATEOAS)	Flask JSON responses	Express res.json()
12	Security Fundamentals	Securing Web Application, Simple User based security implementation, discussed Authentication vs authorization, TLS/SSL handshake, OWASP Top 10, principle of least privilege	Flask-Login, CSRF	Passport.js , Helmet
13	MVC Web Application-deep dive &	Full-stack synthesis: servlet/JSP + JDBC + security.	Tally with Python alternatives & foundation	Tally with Node alternatives & foundation

14	Introduction to Containers	Docker – introduction and containerization of the application	Same	Same
15	AJAX	A browser technology / pattern that shook the world.	Flask full-stack app	Express full-stack app
16	Exploring Modern browser technologies	Server Sent Event, WebSocket & real time communication using vanilla ES (JavaScript)	Same	Same
17	Java annotation and Add-on for next phase	We need a few topics in Java to cleared before moving on to next phase of the training		
18	OO design patterns using Java	Common Java design patterns		
19	Docker hands-on	A definitive beginners' roadmap into containers using Docker		
20	Decoupled & Hexagonal Architecture	Everyone codes but system design and architecture set the winning track		

15-Day Advanced Course Plan: REST CRUD & Microservices with Spring Boot

Day	Topic	Spring Boot Focus – Elaborated
1	Spring Boot Intro	Auto-config heuristics, starter dependencies, embedded servlet containers, convention-over-configuration
2	Creating HTTP and REST APIs	Annotation-driven routing, @RestController, request mapping semantics, JSON SerDe, handling errors
3	Basic CRUD Operations using RDBMS	Repository abstraction, Need of an ORM framework
4	Advanced CRUD operation	JPA/Hibernate, transactional demarcation, persistence context
4	Data Validation	Bean Validation (JSR 380), constraint annotations, exception translation, defensive programming

5	DTOs & Model Mapping	Data Transfer Objects, ModelMapper, anti-corruption layer, serialization boundaries – advanced JSON features
6	Error Handling	Global exception resolvers, error payload standardization, resilience patterns
7	Consuming REST Applications	Blocking approach - consume REST APIs , considerations for calling secured endpoints, logging outbound communication
8	Security in Spring Boot	Spring Security filter chain, authentication providers, role hierarchies, CSRF protection
9	JWT Authentication	Token issuance, signature verification, stateless sessions, bearer token semantics
10	Microservices Architecture	Service decomposition, bounded contexts, inter-service contracts, distributed systems complexity
11	Spring Cloud Basics	Config server, service registry (Eureka), circuit breakers, centralized configuration
12	Inter-Service Communication	Feign client, REST template, synchronous vs asynchronous calls, service discovery
13	API Gateway	Reverse proxy, request routing, rate limiting, gateway filters, Zuul/Spring Cloud Gateway
14	Monitoring & Logging	Spring Boot Actuator, observability, centralized logging (ELK stack), metrics instrumentation
15	Capstone Project	Microservices orchestration, REST CRUD aggregation, containerization, docker compose support